

BESTYAHSKAYA ZVEROFERMA, Russian Federation

WMO: 244490

Lat: **65.300N**Lon: **124.166E**Elev: **240**StdP: **14.57**Time Zone: **9.00 (E09)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-59.8	-56.8	-62.4	0.1	-59.1	-59.2	0.2	-57.0	10.0	-11.7	9.4	-12.9	0.1	310	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	81.6	64.5	78.2	63.0	74.8	61.2	67.3	77.6	65.2	75.1	62.9	71.9	4.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	88.4	71.9	61.3	81.7	69.9	58.6	74.2	68.3	32.0	78.6	30.4	74.7	28.6	72.2	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	9.7	8.6	-60.9	87.6	4.0	2.3	-63.8	89.2	-66.1	90.6	-68.4	91.9	-71.3	93.5		
			-60.9	70.7	4.0	1.9	-63.8	72.1	-66.1	73.2	-68.4	74.3	-71.3	75.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	-36.6	-29.1	-2.9	21.0	39.1	57.7	59.7	55.9	39.7	16.6	-19.4	-34.6
	DBStd	37.28	12.89	12.40	15.27	11.33	9.01	8.01	6.71	6.69	7.51	11.40	12.42	12.07
	HDD50	13873	2683	2215	1641	870	354	21	6	24	318	1035	2083	2623
	HDD65	18612	3148	2635	2106	1320	805	241	191	287	758	1500	2533	3088
	CDD50	791	0	0	0	0	15	252	308	206	10	0	0	0
	CDD65	55	0	0	0	0	1	22	28	4	0	0	0	0
	CDH74	849	0	0	0	0	9	392	359	89	0	0	0	0
	CDH80	160	0	0	0	0	2	83	71	4	0	0	0	0
	WSAvg	3.6	2.4	2.9	3.4	4.2	4.5	4.1	4.3	3.9	3.6	3.9	2.8	2.9
	PrecAvg	15.2	0.6	0.6	0.7	0.7	1.3	1.8	2.0	2.2	1.7	1.9	1.0	0.7
	PrecMax	21.6	1.3	1.4	2.0	1.6	2.3	3.6	3.9	5.2	3.2	3.9	2.2	1.8
(15) Precipitation	PrecMin	11.7	0.2	0.1	0.1	0.2	0.4	0.3	0.5	0.4	0.4	0.6	0.4	0.4
	PrecStd	2.3	0.2	0.3	0.4	0.4	0.4	0.7	0.8	1.0	0.7	0.7	0.5	0.3
	DB	0.1	6.4	36.8	48.7	73.2	86.0	85.8	80.2	69.3	38.4	18.1	-3.9	
	MCWB	-0.4	5.6	31.5	40.1	52.3	66.5	67.5	64.1	57.4	35.6	16.5	-4.4	
	DB	-6.5	0.8	30.2	44.7	66.4	82.3	81.9	76.6	62.7	35.2	11.7	-7.5	
(21) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% MCWB	-6.9	0.1	26.6	36.9	51.0	63.2	66.4	62.5	52.0	33.3	11.0	-7.8	
	DB	-12.1	-6.6	24.9	41.6	61.4	78.8	78.2	72.9	57.3	33.0	6.3	-10.3	
	5% MCWB	-12.4	-7.2	22.2	34.5	48.3	61.8	64.6	61.7	49.5	31.4	5.7	-10.6	
	DB	-17.6	-10.8	21.0	38.6	56.6	75.1	74.7	69.4	53.1	30.6	-0.9	-14.6	
	10% MCWB	-17.9	-11.3	19.1	32.9	46.1	59.8	62.4	59.8	46.4	28.9	-1.6	-14.9	
	0.4% WB	-0.3	5.3	31.8	40.5	54.9	68.2	70.1	68.1	58.5	36.7	16.3	-4.2	
	MCDB	-0.1	6.1	35.8	47.7	69.6	82.9	79.8	77.0	69.6	37.6	17.3	-3.8	
	2% WB	-7.0	0.3	26.6	37.6	51.3	65.8	68.2	65.1	53.1	33.8	11.1	-7.9	
	MCDB	-6.8	1.0	29.7	43.1	63.7	78.6	78.0	72.5	59.9	34.9	11.6	-7.7	
	5% WB	-12.4	-7.2	22.5	35.3	49.4	63.4	66.3	62.8	50.4	31.7	5.6	-10.6	
(31) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	-12.0	-6.8	24.7	40.3	61.5	75.2	75.6	69.9	56.6	33.0	6.3	-10.2	
	10% WB	-17.8	-11.2	19.1	33.5	46.5	61.1	64.0	61.0	47.7	29.0	-1.3	-14.9	
	MCDB	-17.6	-10.8	21.3	38.0	55.9	73.3	73.6	68.8	52.5	30.4	-0.7	-14.5	
	MDBR	13.1	18.3	23.6	22.8	18.7	22.2	20.5	20.6	18.4	12.4	15.1	12.1	
	5% DB	15.1	18.6	23.6	22.3	29.4	30.4	27.4	28.4	26.8	11.3	15.0	14.7	
(36) Mean Daily Temperature Range	MCWBR	15.0	18.0	20.7	16.3	16.5	14.8	13.9	16.0	17.3	9.2	14.8	14.5	
	MCDBR	15.2	18.5	22.3	19.1	26.0	26.4	22.0	24.1	23.9	9.9	15.3	14.6	
	MCWBR	15.0	17.9	19.8	14.1	15.8	13.6	11.4	14.8	16.2	8.5	15.1	14.4	
	taub	0.186	0.243	0.288	0.334	0.372	0.391	0.480	0.472	0.325	0.261	0.185	0.182	
	taud	1.875	2.064	2.085	2.087	2.260	2.343	2.047	2.013	2.450	2.245	1.897	1.881	
(42) Clear-Sky Solar Irradiance	Ebn at Noon	94	203	245	260	263	260	226	212	239	196	89	21	
	Edh at Noon	7	19	30	39	36	34	45	42	22	16	6	2	
	RadAvg	43	301	880	1565	1743	1864	1704	1213	728	261	83	11	
(45) All-Sky Solar Radiation	RadStd	2	14	38	68	122	138	108	128	83	36	6	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A